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2 December 1966
SUPERSEDING
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MIL-T-0013025B(MU)
31 January 1964
MIL-T-13025A
13 April 1954

MILITARY SPECIFICATION

THICKENER, INCENDIARY OIL, M2

1. SCOPE

1.1 This specification covers one type of incendiary oil thickener.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366	-	Sieves, Standard for Testing Purposes.
UU-P-121	-	Paper, Bond and Writing, White and Colored.
PPP-B-35	-	Bags: Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-D-705	-	Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon).
PPP-P-704	-	Pails: Shipping, Steel (1 Through 12 Gallon).

MILITARY

JAN-I-711	-	Incendiary Oils, Consistency Of; Test Procedures.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 791	-	Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

FSC 1365

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS602 - Test Solvent (For Fuel Thickeners).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-56 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material. The thickener shall be an intimate mixture of 95 ± 1 parts by weight of thickening agent and sufficient approved antiagglomerant to total 100 parts by weight (see 6.6). Sufficient alpha-naphthol shall be added as an inhibitor.

3.1.1 Thickening agent. The thickening agent shall consist of a coprecipitated mixture of basic aluminum soaps of naphthenic, oleic, and coconut fatty acids (see 6.5).

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3.2 Inhibitor content. The thickener shall develop a color equal to or darker than a 0.2 percent alpha-naphthol standard when tested as specified in 4.5.1.

3.3 Moisture. The thickener shall contain no more than 0.8 percent moisture when tested as specified in 4.5.2.

3.4 Granulation. No more than 3 percent of the material shall be retained on a No. 20 standard sieve and no more than 30 percent shall pass through a No. 100 standard sieve when tested as specified in 4.5.3.

3.5 Sulfate. The thickener shall contain no more than 3.0 percent sulfate when tested as specified in 4.5.4.

3.6 Gel formation. All gels, prepared and tested as specified in 4.5.5, shall have a vortex time of 30 to 180 seconds.

3.7 Gel consistency. All gels, prepared and tested as specified in 4.5.6, shall have a consistency of 300 to 500 grams (g) after storage at 77° ± 2°F and 250 to 450 g after storage at 150° ± 2°F. The average consistencies of the two samples shall differ no more than 80 g for the same temperature.

3.8 Bulk density. The thickener shall have a bulk density of 0.30 to 0.65 g per cubic centimeter (cc) when tested as specified in 4.5.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Lotting. A lot shall consist of the thickener produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. If the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.3 Sampling.

4.3.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.2 For test. Sampling shall be conducted in accordance with MIL-STD-105, level S-2. A one-pound specimen shall be removed from each container in the sample and placed in a clean dry container labeled to identify the lot and the container from which it was taken. A composite specimen shall be made with equal portions from each specimen except one.

4.4 Inspection provisions.

4.4.1 For examination. Sample containers shall be examined for compliance with level A preparation for delivery requirements in accordance with the classification of defects and MIL-STD-105.

4.4.2 For test. The composite and the individual specimens shall be tested in accordance with 4.5. The acceptance number is 0.

4.4.3 Classification of defects.

(a) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Container not as specified
102	Polyethylene liner damaged or torn
103	Mixing instructions incorrectly packaged, missing, or improperly located
104	Marking missing, incorrect, or illegible
105	Container closure inadequate

4.5 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

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4.5.1 Inhibitor content.

4.5.1.1 Mixed solvent. Mix by volume: five parts iso-octane, five parts toluene, and three parts absolute ethanol.

4.5.1.2 Reagent. Dissolve 4.0 g sodium bromide and 0.10 g cupric chloride in 10 milliliters (ml) water, and store in a stoppered brown bottle. Prepare fresh reagent weekly, and shake thoroughly before use.

4.5.1.3 Alpha-naphthol standard solution. Dissolve 0.10 g of alpha-naphthol in 100 ml of the mixed solvent (see 4.5.1.1). Prepare a fresh solution every two weeks and store in a stoppered brown bottle.

4.5.1.4 Procedure. Weigh 0.5 g of the sample into a stoppered 50-ml flask. Add 25 ml of mixed solvent (4.5.1.1) and shake until the sample is in solution or uniformly suspended in the solvent. Pipet 1 ml of the reagent (4.5.1.2) into the flask and shake for 3 minutes. Pour the contents of the flask into a clean and dry 6- by-5/8-inch test tube. Stopper the tube and allow to stand until the turbidity decreases to a minimum (approximately 5 minutes). Next, prepare a color standard by dissolving 0.5 g of alpha-naphthol-free thickener in 25 ml of mixed solvent (4.5.1.1). Pipet 1.0 ml of standard solution (4.5.1.3) and 1 ml of reagent (4.5.1.2) into the flask and shake for 3 minutes. Pour the contents of this flask into a test tube as above, stopper, and allow to stand until the turbidity decreases to a minimum. Compare the colors of the clear supernatant liquid in the two test tubes.

4.5.2 Moisture. Weigh out rapidly, to the nearest decigram, approximately 25 g of the sample and immediately disperse in 500 $\pm$ 10 ml of benzene in a round bottom liter flask. After the sample has been dispersed (approximately 5 minutes), determine the moisture content in accordance with method 3001.5 of Fed. Std. 791 with the following exceptions and additional details:

(a) Use the entire 500 ml of suspension instead of the 100 ml specified.

(b) Heat the flask by means of a liquid bath or an electric heating mantle. Maintain the temperature of the bath or heating mantle at approximately 80°C during the distillation period.

(c) Distill for 2 hours. Wash down the condenser tube with three 10-ml portions of benzene, scrubbing the tube at the same time with a stiff brush to loosen water droplets. Resume the distillation for an additional 1/2 hour; then repeat the washing and scrubbing.

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Remove the condenser and loosen the water droplets adhering to the side wall of the trap by policing down with a glass rod equipped with a rubber policeman. Form a good meniscus on the water collected in the bottom of the trap by wiping the surface of the trap wall at the interface level, exerting a firm pressure on the trap wall. Free any droplets of benzene caught in the water phase with the policeman. Remove water adhering to the policeman by twirling the policeman in the upper benzene phase.

(d) Protect the top of the condenser tube by a calcium chloride tube or equivalent, except during the rinsing and scrubbing operations.

(e) At all times during the distillation, maintain the heating medium at a level about 1/4 inch below the level of the suspension in the liter flask.

(f) Make a blank determination by subjecting 500 ml of benzene to the same treatment as the suspension and correct all determinations accordingly.

(g) Determine the efficiency of recovery by similarly treating 500 ml of benzene to which 0.50 ml of water has been added. If, after correction for the blank, the apparatus fails to recover 0.50 $\pm$ 0.05 ml of water, tighten all connections and repeat the test. No determination shall be made with apparatus which has not demonstrated satisfactory recovery.

(h) Ground glass connections may be used in the apparatus in place of the tight-fitting corks. Care shall be exercised to prevent the entrance of atmospheric moisture by condensation on such joints.

(i) Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100\ (A-B)}{C}$$

where:
A = milliliters of water collected,
B = blank, and
C = grams of sample.

4.5.3 Granulation. Nest a No. 20 sieve in a No. 100 sieve conforming to RR-S-366 and place a receiving pan at the bottom. Place 50.0 g of the sample in the top sieve, cover, and assemble in a mechanical shaker geared to produce 300 $\pm$ 15 gyrations and 150 $\pm$ 10 taps of the striker per minute. Vibrate for 5 minutes with the tapper in operation. Weigh the portions retained on each sieve and calculate the amounts on a percentage basis.

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4.5.4 Sulfate. Weigh to the nearest milligram approximately 5 g of the sample into a 400-ml beaker. Add 25 ml of concentrated hydrochloric acid and heat to effect solution. Dilute with water to approximately 200 ml and add 10 ml of bromine water. Boil until the solution clears and allow to cool. When cool, filter through No. 40 filter paper and collect the filtrate in a 600-ml beaker. Wash the residue three times and add the washings to the filtrate. Heat the filtrate to boiling and add dropwise 25 ml of 15 percent barium chloride solution and 5 drops of 1 percent agar-agar solution. Allow to cool to room temperature and filter through a tared Gooch crucible. Ignite for approximately 1 hour at a dull red heat (600°C) in a muffle furnace. Cool to room temperature in a desiccator and weigh. Calculate the percent sulfate as follows:

$$\text{Percent sulfate} = \frac{41.15A}{B}$$

where: A = grams of $BaSO_4$, and
B = grams of sample.

4.5.5 Gel formation.

4.5.5.1 Apparatus. The apparatus for gel preparation shall consist of the following:

(a) A mechanical stirrer rotating at a speed of 300 ± 10 revolutions per minute. The stirrer shall be fabricated of a glass or corrosion-resistant metal rod, 3/8 inch in diameter, with four vanes of the same material as the rod, 7/8 by 7/8 by 1/8 inch, set at right angles and with the faces parallel to the axis of the rod.

(b) A square, quart-size, mason-type jar approximately 3-1/2 by 3-1/2 by 7 inches.

(c) Storage containers of the type described in method 100 of JAN-I-711.

(d) Constant temperature baths (air or liquid) capable of maintaining the specified test temperatures.

(e) A weighing room maintained at 77° ± 2°F and a maximum relative humidity of 50 percent.

4.5.5.2 Procedure. Place 281.4 ± 0.2 g of test solvent conforming to MS602 into the glass jar and immerse the stirrer in the solvent to between 1 and 1-1/2 centimeters (cm) above the bottom of the jar. Make a

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mark on the stirrer shaft 1.0 cm below the surface of the solvent. Start the stirrer, adjust the temperature to 77° ± 2°F, and maintain the temperature throughout the test. Add 18.6 ± 0.1 g of the sample. Record the time of addition of the sample and the time at which the gel covers the mark on the stirrer shaft. Report the elapsed time as the vortex time.

4.5.6 Gel consistency. Using the apparatus specified in 4.5.5.1, prepare four gels from each sample by stirring 18.5 ± 0.1 g of sample into 281.5 ± 0.1 g of test solvent conforming to MS602 for each gel and place them in separate storage containers. Maintain the temperature of the solution during stirring at 77° ± 2°F. Condition two gels from each sample for 22 ± 2 hours at 77° ± 2°F and the remaining gels for 22 ± 2 hours at 150° ± 2°F. Determine the consistency of each gel in accordance with method 100 of JAN-I-711. If the consistencies of the individuals within each pair do not agree within 40 g, repeat the procedure.

4.5.7 Bulk density. Assemble the apparatus as shown in figure 1, mounting the funnel 1-1/2 inches above the measuring cup. Place 500 cc of the sample in any suitable container. Close the small end of the funnel. Quickly open the bottom of the funnel and allow the sample to flow freely into the cup. Scrape off the excess with a straight edge. Weigh the material in the cup to the nearest 0.1 g and calculate the weight in grams of 1 cc. The volume of the cup is 400 cc.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. Twenty ± 1/4 pound of thickener shall be packaged in a bag conforming to PPP-B-35, type III, class 6, style B, bag number P55. The bag shall have approximate dimensions of 12 by 4 by 24 inches.

5.2 Packing, level A.

5.2.1 Twenty-pound quantity. Twenty ± 1/4 pound of thickener shall be packed in a 5-gallon pail conforming to PPP-P-704, type II, class 3. Mixing instructions conforming to 5.3.1 shall be inserted in a waterproof envelope, 4-1/4 by 5-3/4 inches. The envelope shall be affixed with suitable adhesive to the inside of the cover.

5.2.2 Hundred-pound quantity. Five bags of thickener, packaged as specified in 5.1, shall be packed in a drum conforming to PPP-D-705, type V. Mixing instructions conforming to 5.3.1 shall be inserted in a waterproof envelope, 4-1/4 by 5-3/4 inches. The envelope shall be affixed with suitable adhesive to the inside of the cover.

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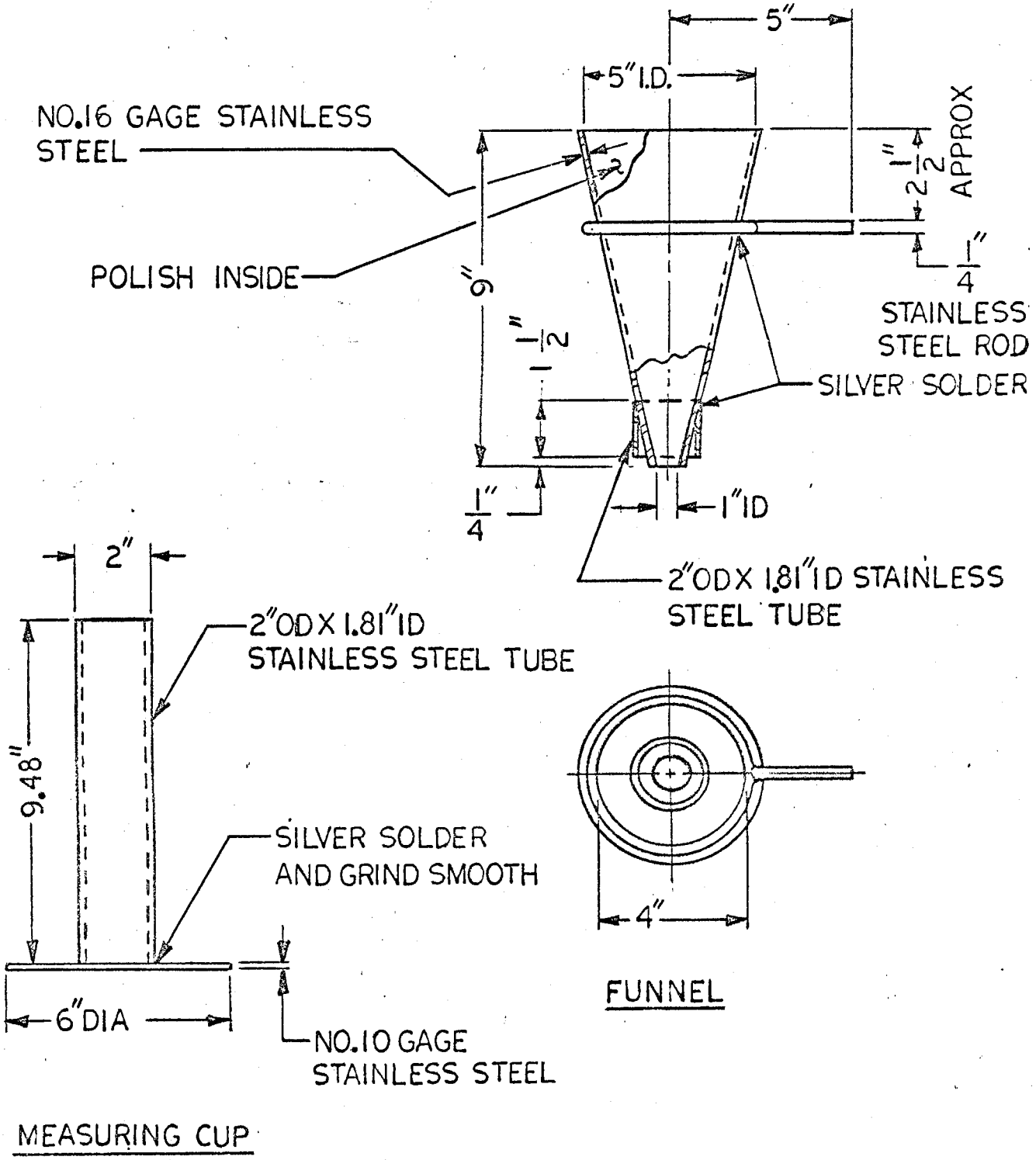


Fig. 1 — Bulk density apparatus

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5.3 Marking. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, the tops of each container shall be marked as follows in 2-inch letters of a contrasting color:

DO NOT OPEN EXCEPT FOR USE OR INSPECTION

MIXING INSTRUCTIONS INSIDE

BULK DENSITY, grams/cc _____*

*To be inserted by manufacturer.

5.3.1 Mixing instructions. Mixing instructions conforming to figure 2 shall be printed with black printer's ink on 8- by 10-1/2-inch paper conforming to UU-P-121, type I or II.

6. NOTES

6.1 Intended use. This material is intended for use in the preparation of thickened incendiary oils for fire bombs.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

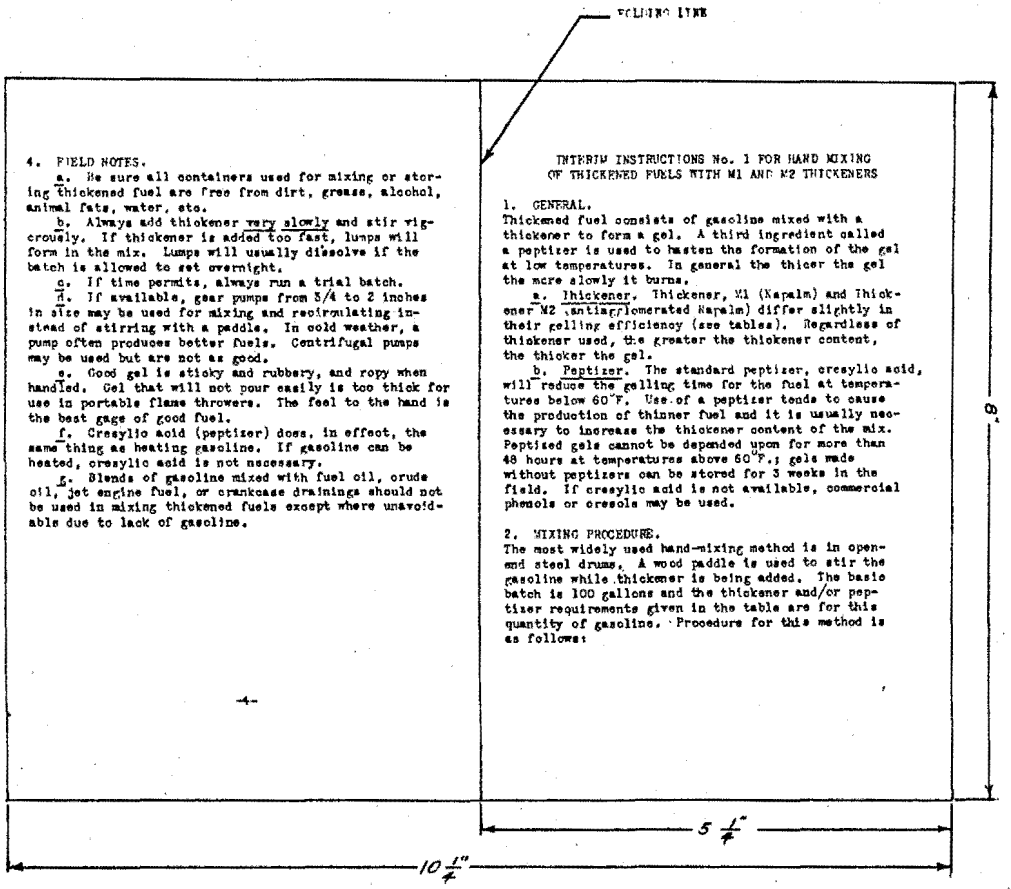
(b) Size pack required.

6.3 Batch. A batch is that quantity of material manufactured by some unit chemical process or subjected to some physical mixing operation to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.5 Thickening agent. Thickener conforming to MIL-T-589 has been found satisfactory for use in the manufacture of Thickener, M2.

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OUTSIDE VIEW BEFORE FOLDING

4. FIELD NOTES.

a. Be sure all containers used for mixing or storing thickened fuel are free from dirt, grease, alcohol, animal fats, water, etc.

b. Always add thickener very slowly and stir vigorously. If thickener is added too fast, lumps will form in the mix. Lumps will usually dissolve if the batch is allowed to set overnight.

c. If time permits, always run a trial batch.

d. If available, gear pumps from 3/4 to 2 inches in size may be used for mixing and recirculating instead of stirring with a paddle. In cold weather, a pump often produces better fuels. Centrifugal pumps may be used but are not as good.

e. Good gel is sticky and rubbery, and ropy when handled. Gel that will not pour easily is too thick for use in portable flame throwers. The feel to the hand is the best gage of good fuel.

f. Cresylic acid (peptizer) does, in effect, the same thing as heating gasoline. If gasoline can be heated, cresylic acid is not necessary.

g. Blends of gasoline mixed with fuel oil, crude oil, jet engine fuel, or crankcase drainings should not be used in mixing thickened fuels except where unavoidable due to lack of gasoline.

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INTERIM INSTRUCTIONS No. 1 FOR HAND MIXING OF THICKENED FUELS WITH M1 AND M2 THICKENERS

1. GENERAL.

Thickened fuel consists of gasoline mixed with a thickener to form a gel. A third ingredient called a peptizer is used to hasten the formation of the gel at low temperatures. In general the thicer the gel the more slowly it burns.

a. Thickener. Thickener, M1 (Napalm) and Thickener M2 (antiagglomerated Napalm) differ slightly in their gelling efficiency (see tables). Regardless of thickener used, the greater the thickener content, the thicker the gel.

b. Peptizer. The standard peptizer, cresylic acid, will reduce the gelling time for the fuel at temperatures below 60°F. Use of a peptizer tends to cause the production of thinner fuel and it is usually necessary to increase the thickener content of the mix. Peptized gels cannot be depended upon for more than 48 hours at temperatures above 60°F.; gels made without peptizers can be stored for 3 weeks in the field. If cresylic acid is not available, commercial phenols or cresols may be used.

2. MIXING PROCEDURE.

The most widely used hand-mixing method is in open-end steel drums. A wood paddle is used to stir the gasoline while thickener is being added. The basic batch is 100 gallons and the thickener and/or peptizer requirements given in the table are for this quantity of gasoline. Procedure for this method is as follows:

a. Above 60°F.

(1) Measure gasoline and pour into open-top steel drums or other containers.

(2) Stir while adding thickener (see tables for amounts).

(3) Continue stirring for about 5 minutes after all the thickener has been added.

(4) Stir for 1 minute every 20 minutes until a suitable gel is formed.

(5) A good gel can be expected in 1 or 2 hours. The gel will improve if allowed to set 2 or more hours.

b. Below 60°F. At these temperatures the procedure is the same as for temperatures above 60°F, except that the amount of thickener is increased and peptizer is added (see tables). After completing steps (1), (2), and (3) above, proceed as follows:

(4) Add peptizer and stir for 5 minutes.

(NOTE: At temperatures below 30°F., peptizer should be dissolved in an equal volume of gasoline before addition.)

(5) Stir for 1 minute every 20 minutes until a suitable gel is formed.

(6) A reasonably good gel can be expected in 4 hours. Gels usually improve if allowed to age 12 to 18 hours.

3. THICKENED FUEL FORMULAS.

Proportions recommended in tables are on the basis of 100 gallons of fuel; changes in quantity of fuel prepared should be in direct ratio to listed amounts. These, however, are to be used as guides only and should be modified as necessary. Due to unpredictable types, qualities, and ages of gasoline, small trial batches are recommended before attempting large quantities.

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MIXING CHART

Gasoline Temperatures	Portable flame thrower		Mechanized flame thrower		Fire bomb	
	Thickener	Peptizer	Thickener	Peptizer	Thickener	Peptizer
	M1 THICKENER					
Above 60°F.	26-1/4	0	40	0	40	0
40 to 60°F.	26-1/4	2-1/2	40	3-1/2	40	5
20 to 40°F.	26-1/4	3-3/4	40	5	50	10
0 to 20°F.	26-1/4	7-1/2	50	7-1/2	60	15
-15 to 0°F.	26-1/4	10	50	8-3/4	70	17-1/2
	M2 THICKENER					
Above 60°F.	20	0	40	0	40	0
40 to 60°F.	25	2-1/2	40	3-1/2	40	5
20 to 40°F.	25	3-3/4	40	5	50	10
0 to 20°F.	25	7-1/2	50	7-1/2	60	15
-15 to 0°F.	25	10	50	8-3/4	70	17-1/2

NOTES: All quantities in pounds per 100-gallon batch.
Thickener, 1 weighs 3-1/4 pounds per gallon.
Thickener, M2 weighs 4 pounds per gallon.
Cresylic acid peptizer weighs 1 pound per pint.
A canteen cup holds 1-1/2 pints.

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INSIDE VIEW BEFORE FOLDING

Figure 2 - INSTRUCTIONS FOR HAND MIXING THICKENED FUELS

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6.6 Antiagglomerants. The following materials have been tested in accordance with Edgewood Arsenal Directive No. 576 and found satisfactory:

(a) Santocel C, manufactured by Monsanto Chemical Company.

(b) Silica gel, grade 974, manufactured by Davison Chemical Co.

(c) Attaclay SF, manufactured by Minerals and Chemicals Philipp Corporation.

(d) Hi-Sil C, manufactured by Columbia-Southern Chemical Corporation.

Custodians:

Army - MU
Navy - AS
Air Force - 70

Review interest:

Navy - AS
Air Force - 70

User interest:

Navy - OS, MC

Preparing activity:

Army - MU (EA)

Project No. 1365-0162

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